GOVERNMENT DEGREE COLLEGE BEERWAH

Understanding Key Concepts in Discrete Mathematics

Sets | Relations | Functions

5th Semester DISCRETE MATHEMATICS

Overview

- Sets
- Types
- Cardinality of a Set
- Set Operations
- Venn Diagrams
- Set Properties and Identities
- Relations and Cartesian Product
- Properties and Types of Relations
- Functions and their Properties
- Composition and Algebra of Functions

Discrete mathematics

- A branch of mathematics that focuses on distinct, separate, and typically countable structures.
- Discrete mathematics provides the tools and theories needed for critical thinking and problem-solving in various areas of computer science, such as algorithms, data structures, databases, and network security.
- The concepts of discrete math are crucial for programming, cryptography, data analysis, and software engineering, as they help build efficient and secure systems.

Key Areas

- **Sets, Relations, and Functions**
- **Logic** and Proof Techniques
- **Combinatorics**
- **Graph Theory**
- Number Theory
- Algorithms and Complexity
- Boolean Algebra

Set

- **Sets** are fundamental discrete structures on which all discrete structures are built .
- A set is an **unordered** | **well defined** collection of **distinct** objects (also called as **elements** or **members** of a set) of **same type**.
- Notation: Sets are usually denoted by **capital letters**, and elements are enclosed in curly braces **{}**.

Example:

- A = {1, 2, 3, 4}
- B = {a,b,c,d}
- C = {2,3,4,1,6,8}
- D = {A,S,D,F}

Representation of Set

Tabular/Roaster Representation

- Here a set is defined by actually listing its elements/members.

Example

A = {1,2,3,4,5} | B = {a,e,i,o,u} | C = {------ -4 , -2, 0, 2, 4, -------}

Set Builder Representation

Here we specify the property which the elements of the set must satisfy

Example

A = {x| x is an odd positive number less than 10}

A = {x | x ∈ English Alphabet && x is Vowel}

B = {x | x ∈ N && x < 5}

C = {x | x ∈ Z && x % 2 = 0}

Cardinality

The total number of elements present in a set.

|A| denotes the number of elements in set A.

Example:

- $A = \{1, 2, 3\}$ $|A| = 3$.

Types of Sets

- **Finite Set**
- **Infinite Set**
- **Subset**
- **Empty Set**
- **Power Set**
- **Universal Set**

Finite Set

- A finite set is a set with a **countable** number of elements.

For example,

- if a set **A={1,2,3,4,5}** then A is finite because we can count that it has exactly five elements.
- Number of students in the class

Infinite Set

- A set contain infinite number of elements is called infinite set
- If the counting of different elements of the set does not come to an end.

Example

- Set of natural numbers. A = { 1,2,3,4,5,6,……………}

Empty Set (Null Set)

- An empty set, or null set, has **no elements** and is denoted by

  $\emptyset$ or $\{\}$.
- It's the unique set with **zero** elements. And its cardinality is 0.
- it's a subset of every set.

Subset

- A subset is a set where all elements in one set are also **found in another set.**
- If A and B are two sets, then A is a subset of B (written as **$A \subseteq B$**)
- if every element of A is also an element of B.

Example

if $A=\{1,2\}$ and $B=\{1,2,3\}$ A then $A \subseteq B$.

Proper Subset

- If A is a subset of B and A is not equal to B then A is said to be a proper subset of B, that is there is at least one element in B which is not in A denoted as $A \subset B$

Power Set

- The power set of a given set A is the **set of all possible subsets** of A, including the empty set and set A itself.
- If A={1,2} the power set of A, denoted P(A) would be:

$$P(A)=\{\emptyset,\{1\},\{2\},\{1,2\}\}$$

- For a set with n elements, the power set has 2^n elements.

Universal Set

- The universal set, often denoted by **U**, is the **set containing all possible elements within a particular context or discussion.**
- It serves as the "universe" for other sets being considered, meaning every other set in that context is a subset of U.
- For example, if we are dealing with sets of integers from 1 to 10 the universal set could be **U={1,2,3,4,5,6,7,8,9,10}**.

General observations

- $\emptyset \subseteq A$, Empty set is a subset of every set.
- $A \subseteq U$, every set is a subset of universal set U.
- $A \subseteq A$, Every set is a subset of itself.

Set Operations

- Set Union
- Set Intersection
- Set Difference
- Set Complement

Set Complement '

- The complement of a set A, denoted **A′** is the set of all elements in the **universal set U that are not in A**.
- It is a unary operator

$$A' = \{ x \mid x \notin A \ \&\ x \in U \}$$

Example:

- **If the universal set U={1,2,3,4,5} and A={2,4}, then: A′={1,3,5}**

Set Union $\cup$

- The union of two sets A and B (denoted as $A \cup B$) is the set of all elements that are either in A or B or both. **And duplicate elements are included once**.
- It is a binary operator.

$$A \cup B = \{ x \mid x \in A \ or \ x \in B \}$$

Example:

If A={1,2,3,4} and B={3,4,5,6} then: A∪B={1,2,3,4,5,6}

Set Intersection ∩

- The intersection of two sets A and B (denoted as $A \cap B$) is the set of all elements that are in both A and B.

$$A \cap B = \{\, x \mid x \in A \text{ and } x \in B \,\}$$

Example:

If A={1,2,3} and B={3,4,5} then $A \cap B$ ={3}

Set Difference – \

- The difference of two sets A and B (denoted as $A - B$ or $A \setminus B$) is the set of elements that are in A but not in B.

$$A - B = \{ x \mid x \in A \ \& \ x \notin B \}$$

Example:

If A={1,2,3} and B={3,4,5} then A−B={1,2} B−A={4,5}

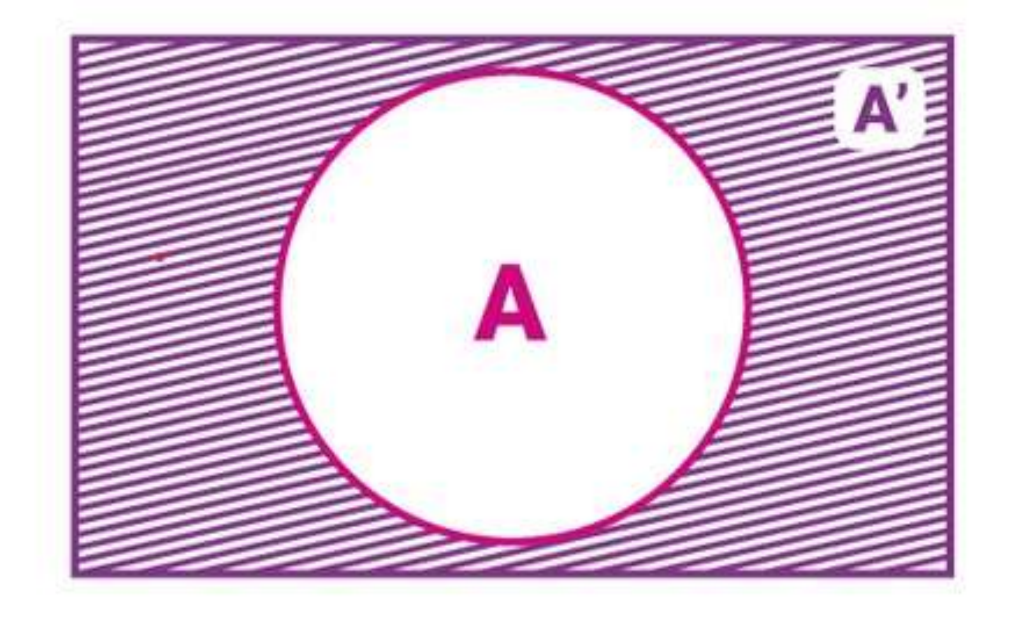
Venn Diagrams

A diagram representing sets and their relationships using circles.

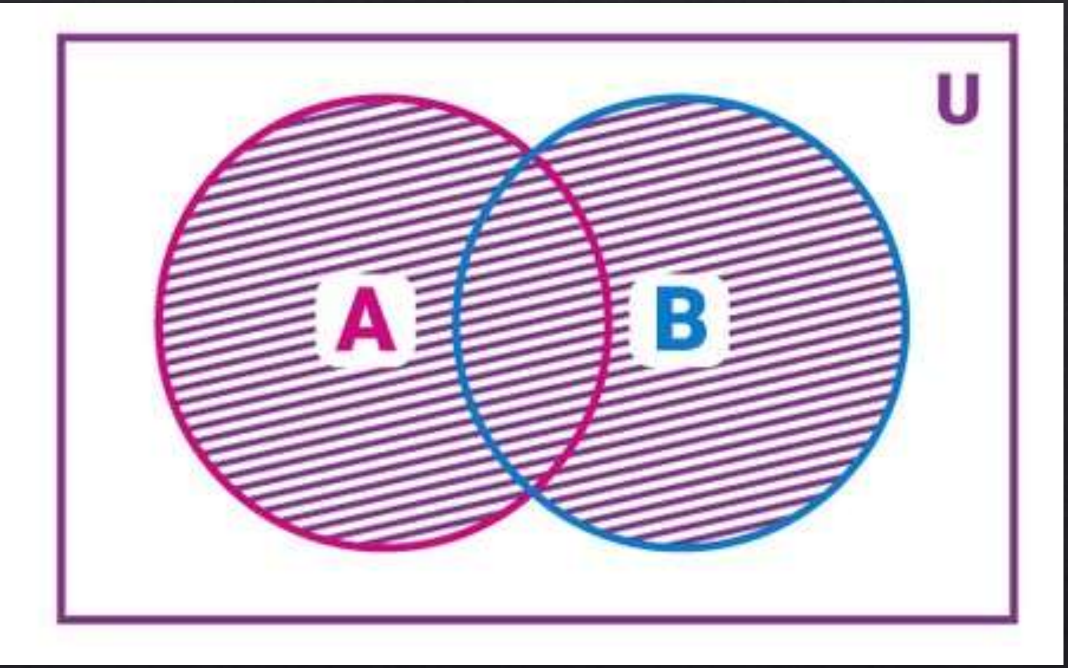
OR

A Venn diagram is a visual representation of mathematical or logical relationships between different sets. They are often used in set theory, logic, probability, statistics, and even everyday reasoning to show the relationships, similarities, and differences between groups.

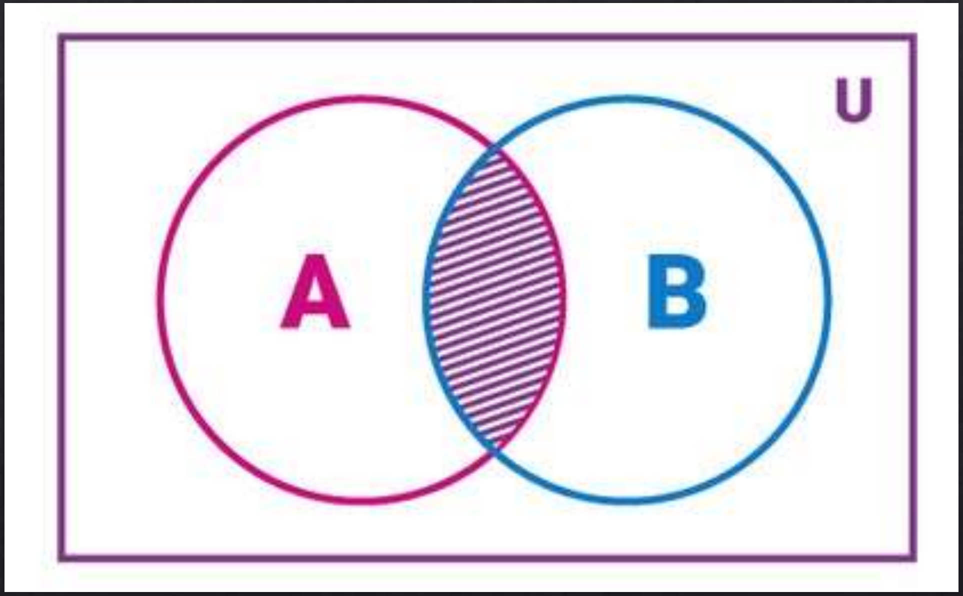
Complement



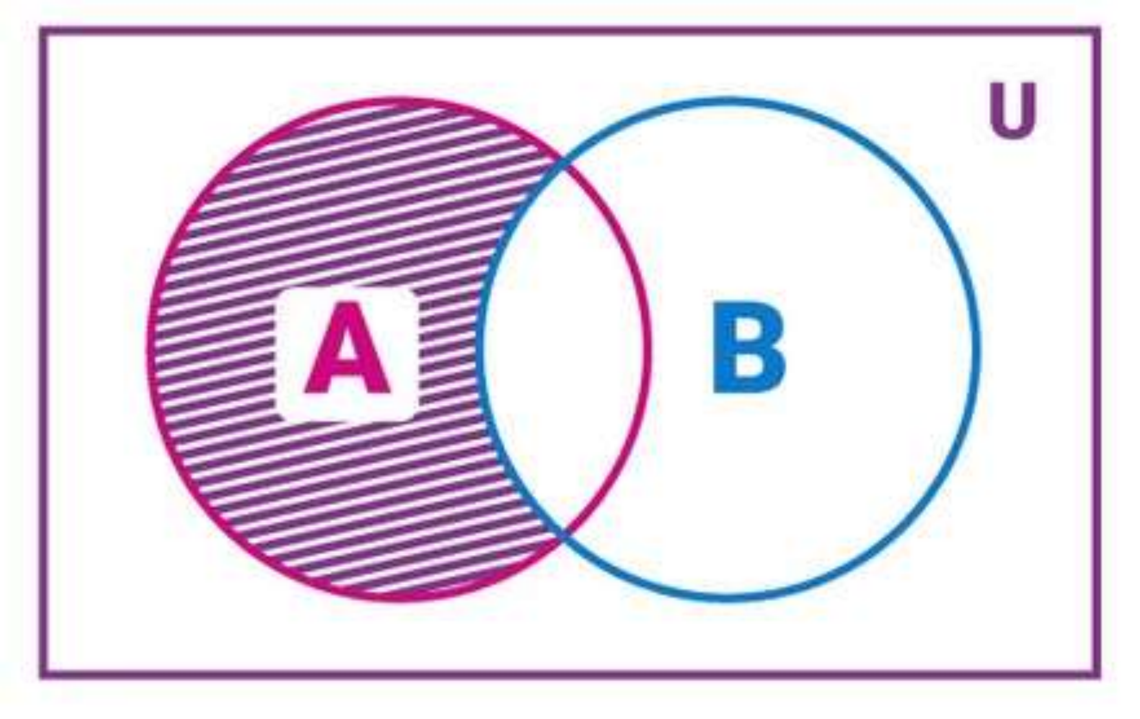
Union



Intersection



Difference



Set Properties and Identities

- **1. Commutative**:

Union $A \cup B = B \cup A$,

Intersection $A \cap B = B \cap A$.

- **2. Associative:**

Union $A \cup (B \cup C) = (A \cup B) \cup C$.

Intersection: $(A \cap B) \cap C = A \cap (B \cap C)$

- **3. Distributive**:

Union $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Intersection $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Set Properties and Identities

- **4. Identity Properties**

$A \cup \varnothing = A$

$A \cap U = A$

- **5. Complement Laws**

$(A \cup B)' = A' \cap B'$

$(A \cap B)' = A' \cup B'$

$(A')' = A$

- **6. Idempotent Laws**

Union: $A \cup A = A$

Intersection: $A \cap A = A$

Set Properties and Identities

- **7. Domination Law**

$A \cup U = U$

$A \cap \emptyset = \emptyset$

- **8. Absorption Laws**

Union with Intersection: $A \cup (A \cap B) = A$

Intersection with Union: $A \cap (A \cup B) = A$

- **9. De Morgan's Laws**

$(A \cup B)' = A' \cap B'$.

$(A \cap B)' = A' \cup B'$

1 Commutative

This property states that the order of sets in a union or intersection operation does not affect the result.

- Union $A \cup B = B \cup A$,
- Intersection $A \cap B = B \cap A$.

Examples

If $A=\{1,2\}$ *and* $B=\{2,3\}$ *then*

$A \cup B=\{1,2,3\}$ *and* $B \cup A=\{1,2,3\}$

$A \cap B=\{2\}$ *and* $B \cap A=\{2\}$

2 Associative

The associative property implies that the grouping of sets in union or intersection does not affect the outcome.

- Union $A \cup (B \cup C) = (A \cup B) \cup C.$
- Intersection: $(A \cap B) \cap C = A \cap (B \cap C)$

Example:

If $A=\{1\}, B=\{1,2\}, C=\{2,3\}$*: then*

$(A \cup B) \cup C=\{1,2,3\}$ *and* $A \cup (B \cup C)=\{1,2,3\}$

3 Distributive

The distributive property allows for the distribution of intersection over union and vice versa.

- Union $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- Intersection $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Example:.

If $A=\{1,2\}, B=\{2,3\}, C=\{1,3\}$ then

$A \cap (B \cup C) = \{1,2\}$ ***and*** $A \cap B) \cup (A \cap C) = \{1,2\}$

4 Identity

The identity properties define operations that leave the original set unchanged.

- Union $A \cup \emptyset = A$
- Intersection $A \cap U = A$

Example:

If $A=\{1,2\}$ *and* $U=\{1,2,3\}$ *then*

$A \cup \emptyset = \{1,2\}$

$A \cap U = \{1,2\}$

5 Complement

Complement laws describe the behavior of sets and their complements within a universal set.

- Union $(A \cup B)' = A' \cap B'$
- Intersection $(A \cap B)' = A' \cup B'$
- Double Complement $(A')' = A$

Example:

If $A = \{1\}$ *and* $U = \{1,2,3\}$ *then*

$A' = \{2,3\}$

$(A')' = A = \{1\}$

6 Idempotent

This law states that performing a union or intersection of a set with itself does not change the set.

- Union: $A \cup A = A$
- Intersection: $A \cap A = A$

Example:

If $A = \{1,2\}$

$A \cup A = \{1,2\}$

$A \cap A = \{1,2\}$

7 Domination

The domination law states that a union with the universal set or an intersection with the empty set results in either the universal set or the empty set, respectively.

- Union $A \cup U = U$
- Intersection $A \cap \varnothing = \varnothing$

Example:

If $A = \{1\}$ and $U = \{1,2,3\}$

$A \cup U = \{1,2,3\} = U$

$A \cap \varnothing = \varnothing$

8 Absorption

The absorption law combines union and intersection, showing that the presence of one operation within another cancels out the other operation.

- Union with Intersection: $A \cup (A \cap B) = A$
- Intersection with Union: $A \cap (A \cup B) = A$

Example:

If $A = \{1,2\}$ *and* $B = \{2,3\}$

$A \cup (A \cap B) = \{1,2\} = A$

$A \cap (A \cup B) = \{1,2\} = A$

9 De Morgan's Laws

De Morgan's laws provide rules for the complement of unions and intersections.

- Complement of a Union $(A \cup B)' = A' \cap B'$.
- Complement of an Intersection $(A \cap B)' = A' \cup B'$

Example:

If $A=\{1\}$ $B=\{2\}$ *and* $U=\{1,2,3\}$

$(A \cup B)'=\{3\}=A' \cap B'$

$(A \cap B)'=\{1,2,3\}=A' \cup B'$

These properties and identities simplify the process of working with sets and provide a foundational understanding of set operations. Each can be applied to solve complex problems in mathematics, computer science, and data analysis.

What is a Relation?

Relation is defined as the relation between two different sets of information.

The relation connects the value of the first set with the value of the second set.

If we are given two sets Set X and Set Y then the relation between them is represented using the ordered pair (a, b) such that, $a \in X$ and $b \in Y$.

or

A relation is a connection between elements of two sets. For instance, a relation from set A to set B is a subset of the Cartesian product $A \times B$, representing pairs of elements where the relation holds

For example

if $A = \{1,2\}$ and $B=\{3,4\}$ a relation R from A to B could be $\{(1,3),(2,4)\}$

Cartesian Product of Two Sets

- Definition: The set of all ordered pairs (a, b) where $a \in A$ and $b \in B$.

Example:

If $A = \{1, 2\}$ and $B = \{x, y\}$,

$A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$.

Representation of Relations

Roaster Notation

- In roaster form, we use ordered pairs to represent the relation.

Example

X = {2, 4, 6} and set Y = {4, 8, 12}.

R = {(2, 4), (4, 8), (6, 12)}

Set Builder Notation

- if a relation between two sets is represented using the logical formula then this type of representation is called the set builder notation.

Example

X = {2, 4, 6} and set Y = {4, 8, 12}

R {(a, b): b is twice of a, a ∈ X, b ∈ Y}

Properties of Relations

Relations have several important properties, often defined based on characteristics of elements within a relation:

- **Reflexive**: A relation R on a set A is reflexive if every element is related to itself. Formally, $\forall a \in A, (a,a) \in R$
- **Symmetric**: A relation R on A is symmetric if whenever $(a,b) \in R$ then $(b,a) \in R$
- **Antisymmetric**: A relation R is antisymmetric if $(a,b) \in R$and $(b,a) \in R$ implies $a=b$
- **Transitive**: A relation R is transitive if $(a,b) \in R$ and $(b,c) \in R$ implies $(a,c) \in R$

Example:

For $A=\{1,2,3\}$ if $R=\{(1,1),(2,2),(3,3),(1,2),(2,3)\}$

this relation is **reflexive** and **transitive** but not symmetric (since (2,1) is missing).

Types of Relations

- Empty Relation
- Reflexive Relation
- Symmetric Relation
- Transitive Relation
- Equivalence Relation

Empty Relation

A relation R on a set A is called Empty if the set A is an empty set, i.e. any relation where no element of set A is not related to the element of set B then it is called an empty relation.

For example,

A = {1, 2, 3} and B = {5, 6, 7}

where, R = {(x, y) where x + y = 22},

then it is an empty relation.

Reflexive Relation

A relation R on a set A is called reflexive if $(a, a) \in R$ holds for every element $a \in A$. i.e. if set $A = \{a, b\}$ then $R = \{(a, a), (b, b)\}$ is reflexive relation.

Example

$A = \{2, 3\}$ then the reflexive relation R on A is,

$R = \{(2, 2), (3, 3)\}$

Symmetric Relation

A relation R on a set A is called symmetric if $(b, a) \in R$ holds when $(a, b) \in R$
i.e. The relation $R = \{(a, b), (b, a)\}$ is a reflexive relation on (a, b)

Example

$A = \{2, 3\}$ then symmetric relation R on A is,

$R = \{(2, 3), (3, 2)\}$

Transitive Relation

A relation R on a set A is called transitive if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$ for all $a,b,c \in A$

Example

set $A = \{1, 2, 3\}$ then transitive relation R on A is,

$R = \{(1, 2), (2, 3), (1, 3)\}$

Equivalence Relations

An **equivalence relation** on a set A is a relation that is **reflexive**, **symmetric**, and **transitive**. Equivalence relations partition a set into **equivalence classes**, where each class contains elements related to each other.

Example:

R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2), (1, 3), (3, 1)}

on set A = {1, 2, 3}

is equivalence relation as it is reflexive, symmetric, and transitive.

Closures of Relations

The closure of a relation is the smallest relation containing the original relation that meets specific properties.

- **Reflexive Closure**: Adds pairs (a,a) for all $a \in A$
- **Symmetric Closure**: Adds pairs (b,a) whenever $(a,b) \in R$.
- **Transitive Closure**: Adds pairs (a,c) if $(a,b) \in R$ and $(b,c) \in R$.

Example:

For $R=\{(1,2),(2,3)\}$

the **transitive closure** would add $(1,3)$

so R becomes $\{(1,2),(2,3),(1,3)\}$.

Functions

What is a Function?

Function is a special type of relation in which the two entities are exclusively related to each other only.

A relation from set A to set B is defined as function if all the elements of set A are related to at least one element of B and no two elements of B are related to a single element of A.

or

A function is a relation between two sets where every element of the first set (domain) is related to exactly one element of the second set (codomain).

It should be noted that **every function is a relation** but **not every relation is a function.**

Notation: $f: A \to B$

indicates that f is a function from set A (domain) to set B (codomain).

Terms Related to Relation and Function

Domain | Codomain | Range

Domain

Domain of Relation or a function is the set of inputs for which the outputs are obtained.

Example,

A = {-1, 0, 1, 2} and B = {0, 1, 2, 3, 4} and set A is related to set B as $a^2 = b$, the set A is the domain of the relation.

Codomain

Codomain is the set of outputs or the image of the relation and function. Codomain may contain exact or more number of elements than the output.

Example

A = {-1, 0, 1, 2} and B = {0, 1, 2, 3, 4} and set A is related to set B as $a^2 = b$,

set B is the codomain.

In set B there is element 3 which is not a perfect square hence it will not have a pre-image.

Range

Range is the set of all outputs which has a pre image. In range all elements are related.

Hence, it has exact number of elements for which relation is defined.

Thus, Range is subset of codomain.

Example

A = {-1, 0, 1, 2} and B = {0, 1, 2, 3, 4} and set A is related to set B

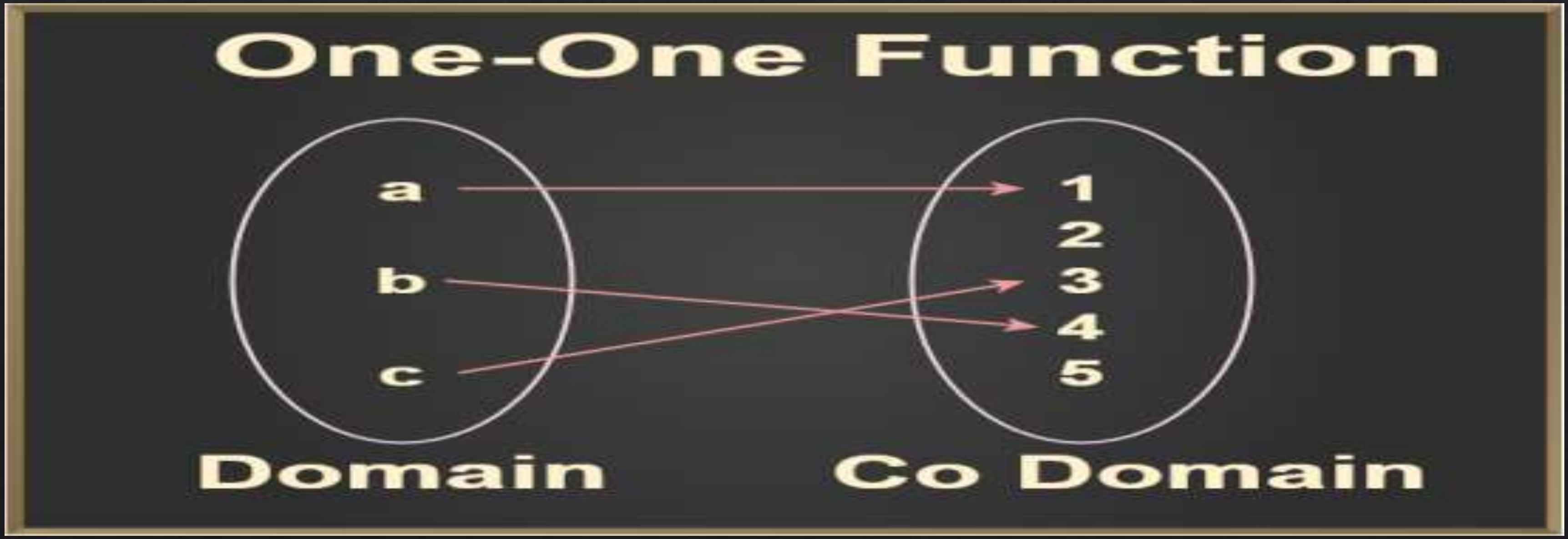
$a^2 = b$, Range is {0, 1, 2, 4}.

Types of Functions

- One-to-One (Injective)
- Many to One
- Onto (Surjective):
- Into
- Bijective:

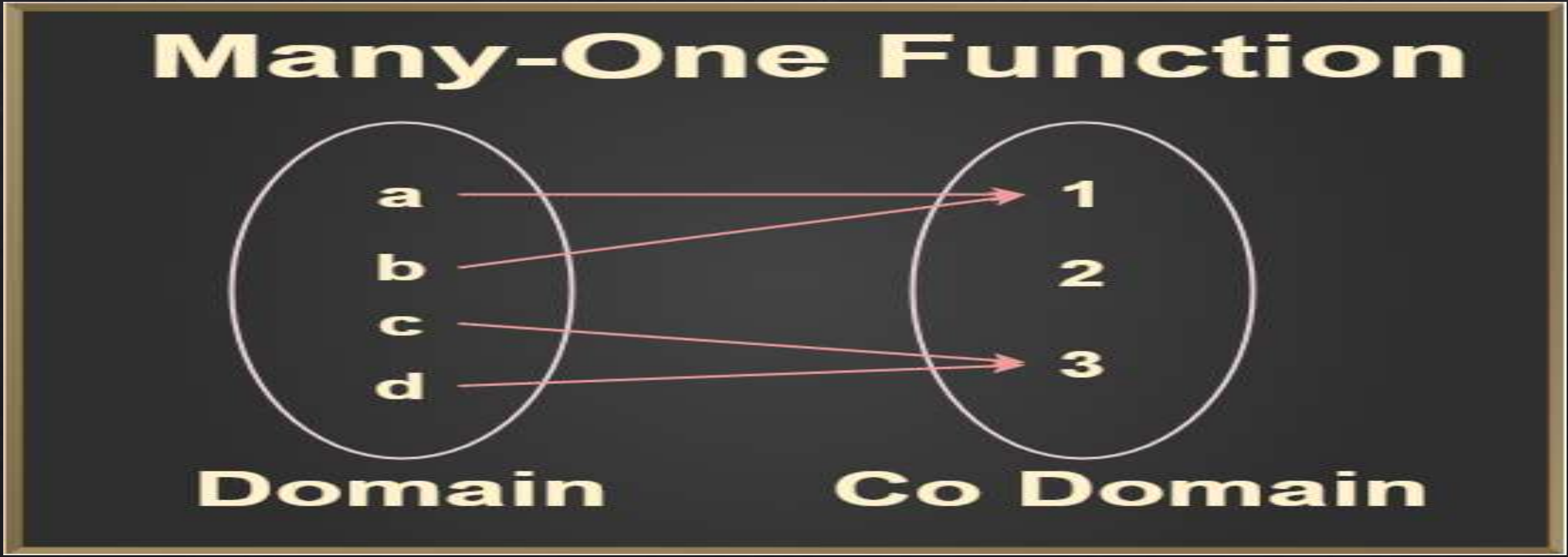
One-to-One

An **injective** function also known as (**injection or one to one** function) is a function that maps distinct elements of its domain to distinct elements of its codomain. In other words, every element of the function's codomain is the image of at most one element of its domain.



Many to One

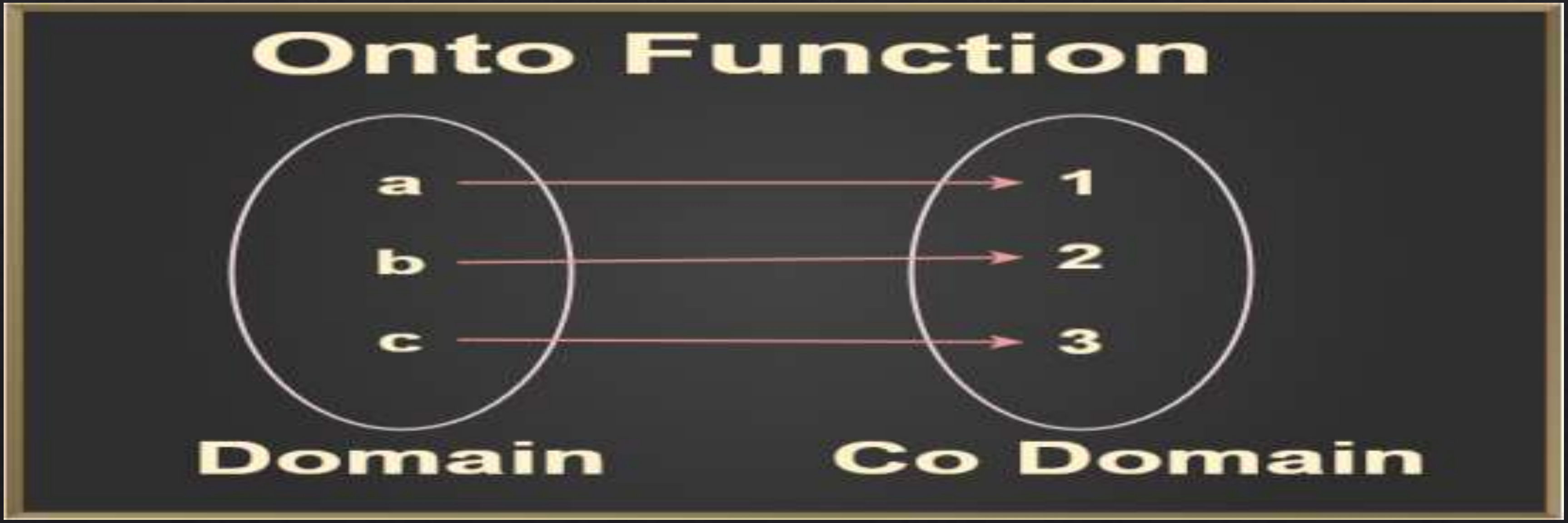
If the function is **not one to one function**, then it should be **many to one function** means every element of the domain **has more than one image** at codomain after mapping.



Onto

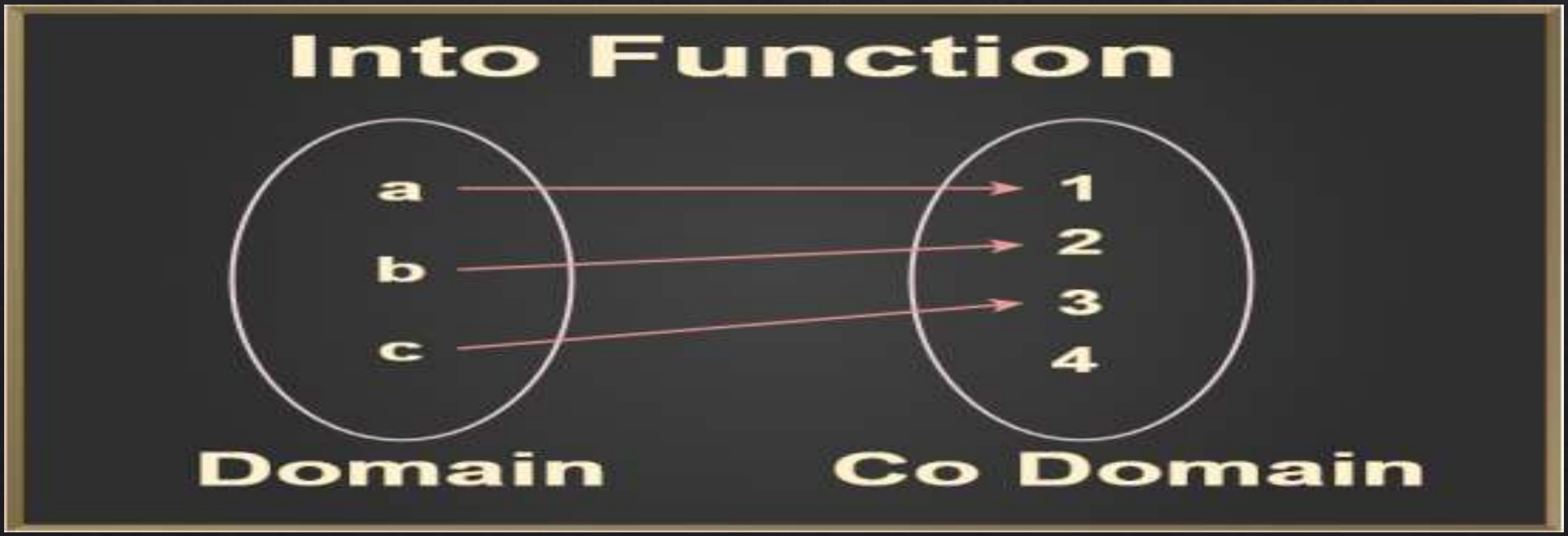
A function f: A → B is said to be **onto** (also known as **Surjective**) if and only if every element of B is mapped by at least one element of A

- The range of functions should be equal to the codomain.
- Every element of B is the image of some element of A.



Into

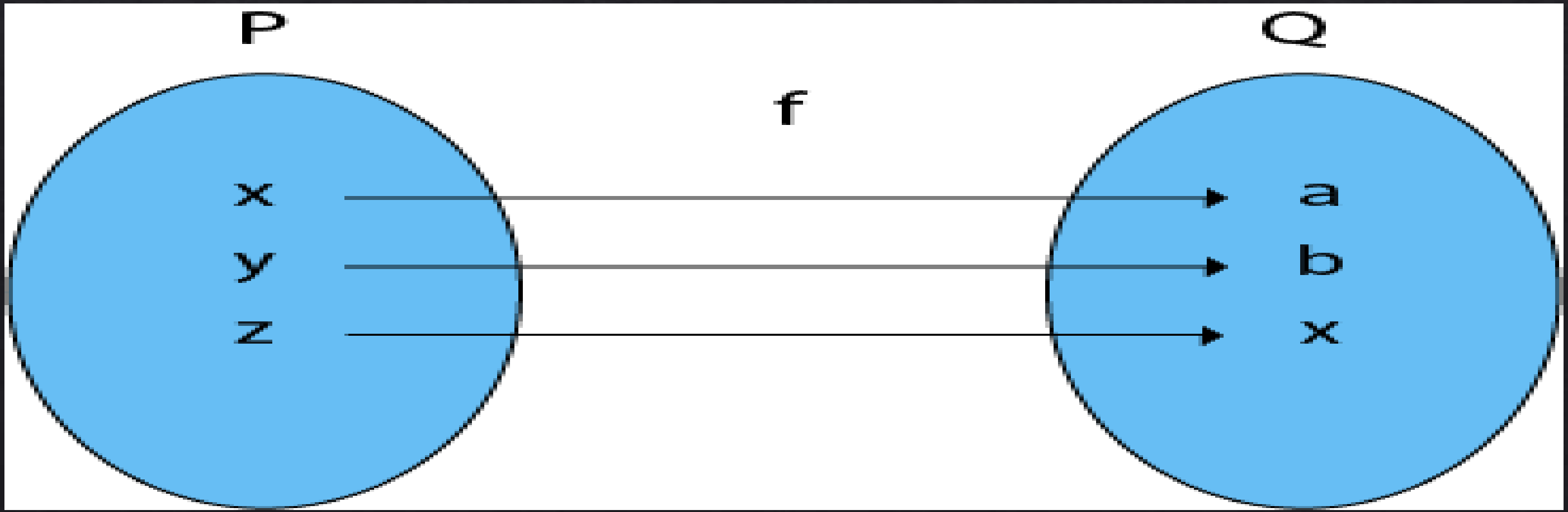
A function f: X → Y is said to be an **into** a function if there exists at least one element or more than one element in Y, which does not have any pre-images in X, which simply means that every element of the codomain are not mapped with elements of the domain.



Bijective

A **Bijective** function is a function between the elements of two sets, where each element of one set is paired with exactly one element of the other set, and each element of the other set is paired with exactly one element of the first set.

There is no unpaired elements.



Composition of Functions

Composition of functions is an operation that takes two functions *f* and *g* and produces a function ***h*** such that ***h(x) = g(f(x)).***

It involves applying one function to the result of another function.

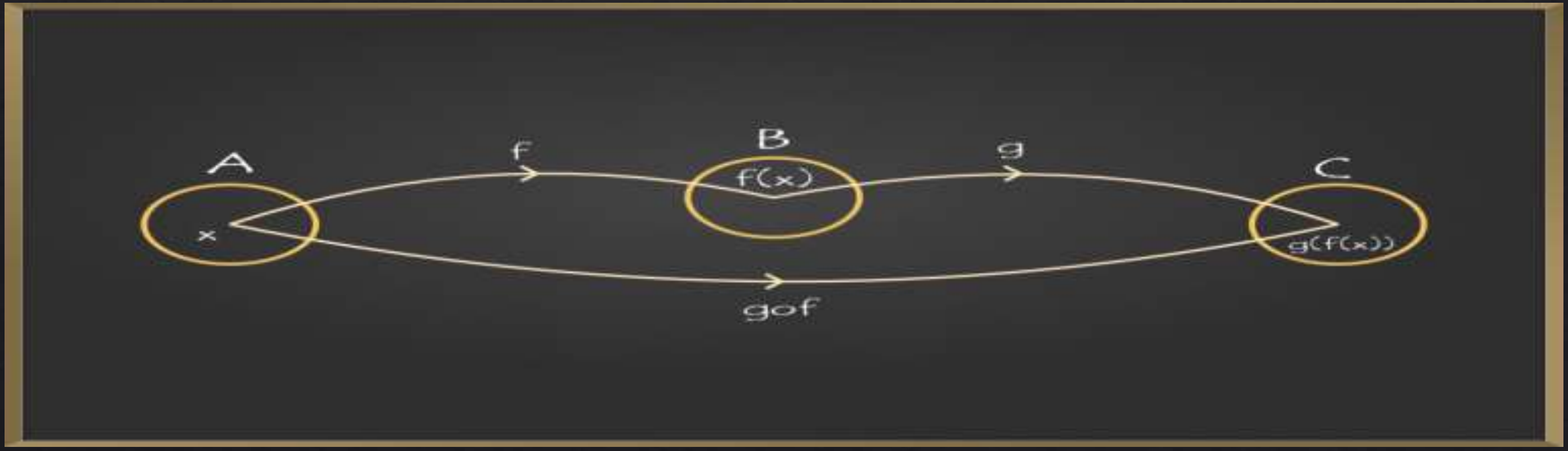
The composition of two functions ***f*** and ***g*** is denoted by

$$f \circ g(x) = f(g(x)) \quad | \quad g \circ f(x) = g(f(x))$$

Example

f = {(1,3), (2,1),(3,4),(4,6)} and g= {(1,5), (2,3),(3,4),(4,1),(5,3), (6,2)} then

$g \circ f$ = {(1,4), (2,5),(3,1),(4,2)}



Algebra of Functions

Algebra of functions talks about the operation performed on of functions like:

Addition | Subtraction | Multiplication | Division

The **sum** of two functions, $(f + g)(x)$ is defined as:

$(f + g)(x) = f(x) + g(x)$

The **difference** of two functions, $(f - g)(x)$ is defined as:

$(f - g)(x) = f(x) - g(x)$

The **product** of two functions, $(f \cdot g)(x)$ is defined as:

$(f \cdot g)(x) = f(x) \cdot g(x)$

The **division** of two functions, $(f / g)(x)$ is defined as:

$(g/f)(x) = g(x) / f(x)$

Summary

We covered:

- 1. Sets and their operations
- 2. Relations and their properties
- 3. Functions and their types
- 4. Composition of functions
- 5. Algebra of functions